

Evaluation Board for the ADBMS6830 16-Channel Battery Stack Monitoring System

FEATURES

Full featured demonstration boards for the ADBMS6830

Built-Features

Demonstrates the High-Performance Measurements of the ADBMS6830

Included 2 isoSPI® ports for reversible isoSPI® support

Daisy-chain capability

Cell and GPIO Accessible via optional connectors

isoSPI® connections via simple DuraClik®

connectors

Compatible Boards

EVAL-ADBMS6822, dual master isoSPI® board

EVAL-SDP-CK1Z, controller board

DC3012A, Legacy ADBMS6830 board

PC software for control and data analysis when used with the Analog Devices, Inc., SDP-K1 microcontroller

EVALUATION BOARD CONTENTS

EVAL-ADBMS6830 isoSPI® daisy-chain slave board
isoSPI® DuraClik® cable

EQUIPMENT NEEDED

EVAL-SDP-CK1Z controller board

EVAL-ADBMS6822 dual master isoSPI® adapter board

SOFTWARE NEEDED

Evaluation software for the ADBMS6830

ADBMS68xx_GUI Windows PC based Graphical User Interface Program

(Request through

https://form.analog.com/form_pages/softwaremodules/SRF.aspx)

GENERAL DESCRIPTION

Evaluation board EVAL-ADBMS6830 is a multicell battery-stack monitor featuring the ADBMS6830, a 16-Channel battery-stack monitor. Multiple boards can be linked through a 2-wire isolated serial interface (isoSPI®) to monitor a long series of cells in a stack. The eval board also features reversible isoSPI enabling a redundant communication path. The PCB, components, and DuraClik® connectors are optimized for Low EMI Susceptibility and Emissions.

EVAL-ADBMS6830 can communicate to a PC by connecting an EVAL-ADBMS6822 Dual Master isoSPI® together with EVAL-SDP-CK1Z (SDP-K1). The SDP-K1 provides a standard SPI interface which can be translated to isoSPI® by EVAL-ADBMS6822 and then connected to a EVAL-ADBMS6830 isoSPI port (J3 or J4 connector). The separate EVAL-ADBMS6822 companion board provides two SPI-isoSPI® channels for reversible operation.

Design files for this circuit board are available.

PERFORMANCE SUMMARY

Table 1. Specifications are at T_A = 25°C

PARAMETER	MIN	TYP	MAX	UNITS
V ⁺ Supply Voltage	11	60	85	V
C _x Voltage	-0.3		85	V
S _{xN} ,S _{xP} Voltage (x=1,2,3)	-0.3		24	V
S _{xN} ,S _{xP} Voltage (x>3)	-0.3		85	V
V _{REG} Supply Voltage	4.5	5	5.5	V
V _{REF1} 1 ST Reference Voltage, No Load	3.0	3.2	3.3	V
V _{REF2} 2 ND Reference Voltage, 5k Load to V-	2.995	3	3.005	V
C _{PIN} Input Range	-2.5		5.5	V
Cell Count			16	

TABLE OF CONTENTS

Features.....	1	EVAL-ADBMS6830 SERIAL INTERFACE OPTIONS.....	4
EVALUATION BOARD Contents.....	1	EVAL-ADBMS6830 OPTIONAL CONNECTIONS	5
Equipment Needed	1	EVAL-ADBMS6822 to EVAL-ADBMS6830 TYPICAL isoSPI	
Software Needed.....	1	CONNECTION	5
General Description.....	1	EVAL-ADBMS6822 to EVAL-ADBMS6830 REVERSE isoSPI	
PERFORMANCE SUMMARY	1	CONNECTION	6
Revision History	2	EVAL-ADBMS6822 to EVAL-ADBMS6830 REDUNDANT	
FUNCTIONAL BLOCK DIAGRAMS	3	isoSPI CONNECTION	7
Evaluation Board Features	4	Evaluation Board Software	9
HARDWARE SETUP.....	4	WHERE TO GET ADBMS68XX_GUI SOFTWARE?.....	9
JUMPERS	4		

REVISION HISTORY

1/2021—Revision 0: Initial Draft Version

7/2021—Revision 1: Updated Draft Version

FUNCTIONAL BLOCK DIAGRAMS

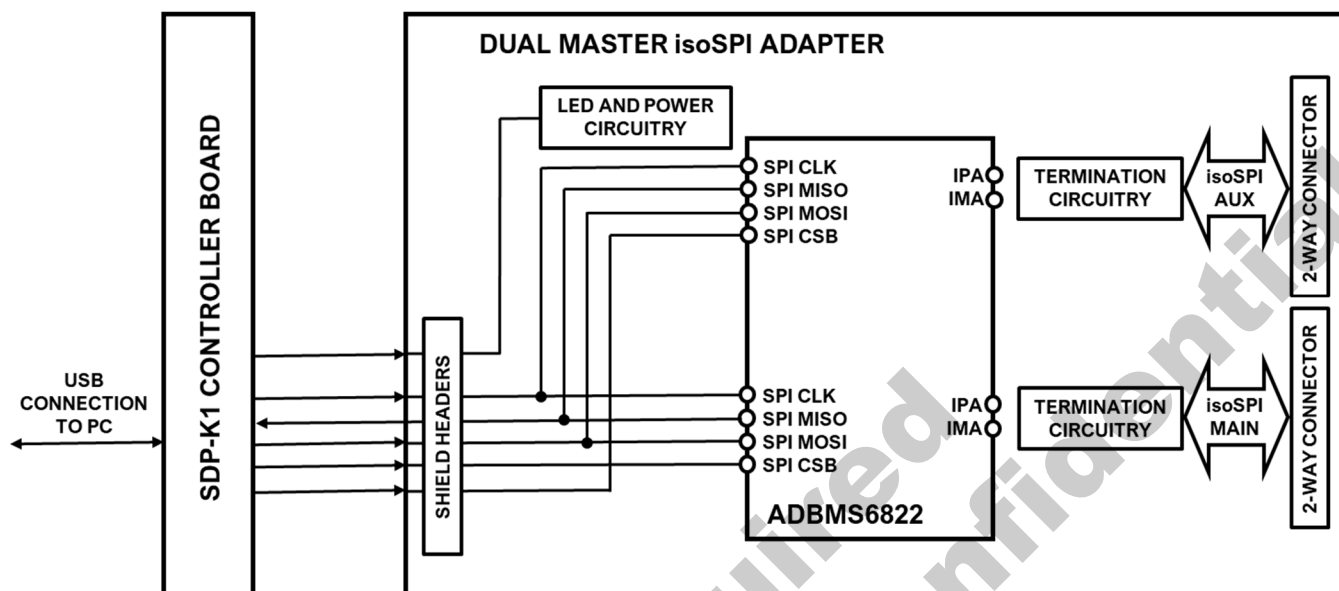


Figure 1. Overview of EVAL-ADBMS6822 Dual isoSPI Master Adapter and EVAL-SDP-CK1Z Interface Board (Both sold separately)

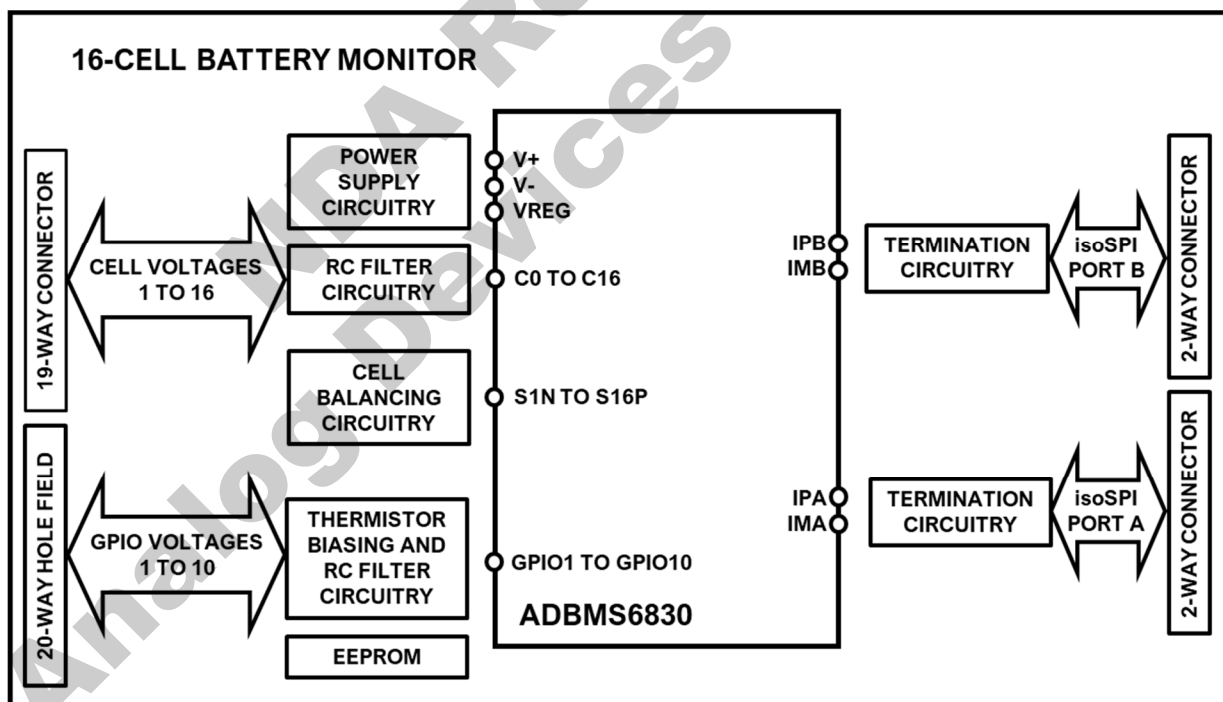


Figure 2. Overview of isoSPI Slave Evaluation Board

EVALUATION BOARD FEATURES

HARDWARE SETUP

Wiring J1 Connector

EVAL-ADBMS6830 board connection is critical; correct wiring must be followed to avoid the risk of damaging the eval board.

When connected to a battery stack, power for the EVAL-ADBMS6830 is provided by the cell group being monitored. To connect the cell group, separate the screw-terminal block section from the J1 connector. Then, insert the cell-voltage connections or resistors into the screw-terminal clamping contacts. These connections provide the power and input stimulus for the battery stack monitor IC.

Cell-voltages are wired to J1 starting from position 1 (most negative potential of the group). Please reference the appropriate eval board J1 connector pinout in Table 1.

Alternatively, resistors can be used to simulate battery cell-voltages. 100Ω 1/2W or equivalent resistors are recommended because 100Ω (or lower values) typically will not induce measurement errors and the 1/2W (or greater rating) will keep the resistor temperatures low preventing power dissipation damage.

EVAL-ADBMS6830 16 Resistor Connection

Carefully connect 16 100Ω resistors onto the screw-terminal block between each CPIN input clamping contact from position 1 to position 17 as shown in Table 2, EVAL-ADBMS6830 J1 Pinout and Figure 3, Resistor Divider String. Provide a stack-equivalent power supply connection to position 17 (positive) and position 1 (negative). The power supply may be adjusted to provide the desired nominal cell-voltage (ex. 52.8V will be 3.3V per cell).

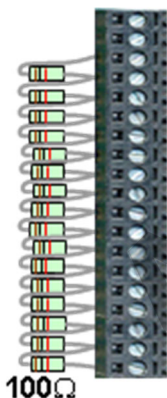


Figure 3. Typical 16-cell Resistor Divider String Connection

Table 2. Pin Designations for the J1 Sixteen Way Connector

Pin No.	Connection
1	C0 (stack-)
2	C1
3	C2
4	C3
5	C4
6	C5
7	C6
8	C7
9	C8
10	C9
11	C10
12	C11
13	C1
14	C13
15	C14
16	C15
17	C16 (stack+)

JUMPERS

There are no jumpers to configure on the EVAL-ADBMS6830.

EVAL-ADBMS6830 SERIAL INTERFACE OPTIONS

isoSPI is the only communication option to EVAL-ADBMS6830. Due to the custom EMI optimized isoSPI cable with DuraClik® connectors, it's highly recommended to use EVAL-ADBMS6822 Dual Master isoSPI demo board or equivalent for easy plug-and-play operation. The EVAL-ADBMS6822 Dual Master isoSPI demo board can be connected as a typical single-ended isoSPI bus master or to both ends of a reversible configuration with two isoSPI bus masters. Refer to demo manual EVAL-ADBMS6822 for usage details.

EVAL-ADBMS6830 OPTIONAL CONNECTIONS

“GPIO” Optional Header J5

This double row of thru-holes (hole field) can be used to as test points for GPIO voltages or can be wired with individual V-/GND connections with each GPIO.

Table 3. Pin Designations for the J5 Twenty Way Connector

Pin No.	Connection
1	GPIO1
2	GPIO2
3	GPIO3
4	GPIO4
5	GPIO5
6	GPIO6
7	GPIO7
8	GPIO8
9	GPIO9
10	GPIO10
11	V-
12	V-
13	V-
14	V-
15	V-
16	V-
17	V-
18	V-
19	V-
20	V-

“IREG” Optional Header J2

This pair of thru-holes can be used to measure “IREG” with an ammeter. Use of an ammeter bypasses the D2 LED (VREG active) which allows accurate measurement of IREG current. Add an ammeter between J2 pins 1 and 2 to measure IREG.

“SUPPLY” Optional Header J6

This pair of thru-holes can be used to insert or measure supplied voltage with a power supply or voltmeter, respectively. To furnish power separately from the cell stack, remove R55 and power board using J6.

EVAL-ADBMS6822 to EVAL-ADBMS6830 TYPICAL isoSPI CONNECTION

A typical isoSPI connection begins with the isoSPI Master connected to the first (or “bottom”) EVAL-ADBMS6830. Additional EVAL-ADBMS6830¹ boards can be daisy-chained onto the isoSPI bus. Communication begins from the first (or “bottom”) EVAL-ADBMS6830 then to the next “upper” EVAL-ADBMS6830 and, finally, to the last (or “top”) EVAL-ADBMS6830.

Figure 4 shows the following connections for two boards on a stack interfaced to a PC:

1. Connect the EVAL-ADBMS6822 Dual Master isoSPI demo board to the EVAL-SDP-CK1Z (referred to hereafter as SDP-K1 and labelled as such).
 - a. mate the EVAL-ADBMS6822 board directly onto the ‘shield’ headers of the SDP-K1.
 - b. Set the P14 jumper of the SDP-K1 to the 3.3V position.

2. Connect a USB cable from the PC USB port to the SDP-K1 P2 connector.
3. Connect the EVAL-ADBMS6822 to the EVAL-ADBMS6830. This EVAL-ADBMS6830 is the first (or “bottom”) board of the stack.
 - a. Connect a 2-wire twisted-pair patch cable² from the EVAL-ADBMS6822 J1 MAIN DuraClik® connector to the “bottom” EVAL-ADBMS6830 J3 isoSPI A DuraClik® connector.
4. Connect or daisy-chain the EVAL-ADBMS6830 to another EVAL-ADBMS6830 in isoSPI mode. This EVAL-ADBMS6830 is the last (or “top”) board of a two-board stack. More EVAL-ADBMS6830 “upper” boards can be daisy-chained together in the same manner.
 - a. Connect a 2-wire twisted-pair patch cable² from the “bottom” EVAL-ADBMS6830 J4 isoSPI B DuraClik® connector to the next “upper” or “top” EVAL-ADBMS6830 J3 isoSPI A DuraClik® connector.
5. CAUTION! Prevent damage to the EVAL-ADBMS6830. Refer to Table 1 and confirm that the cell-voltage

connections to the screw-terminal block matches the EVAL-ADBMS6830 J1 pinout.

- a. Plug the screw-terminal block(s) into the J1 cell-voltage connector(s).
6. A Windows PC based Graphical User Interface is available. Refer to the ADBMS68xx_GUI Software Manual to properly setup the PC with the ADBMS68xx_GUI software to allow communication to the EVAL-ADBMS6830 boards. Refer to the Where To Get The ADBMS68xx_GUI? of this demo manual to request this software.

¹ DC3012A, Legacy DC3012A are compatible with EVAL-ADBMS6830.

² CAUTION! The 2-wire twisted-pair patch cable with the DuraClik® end plugs have 1mm thick center locking tabs on the wiring side that must be pressed down to release from the DuraClik® receptacles. Failure to do so may damage the cable and prevent board-to-board isoSPI communication.

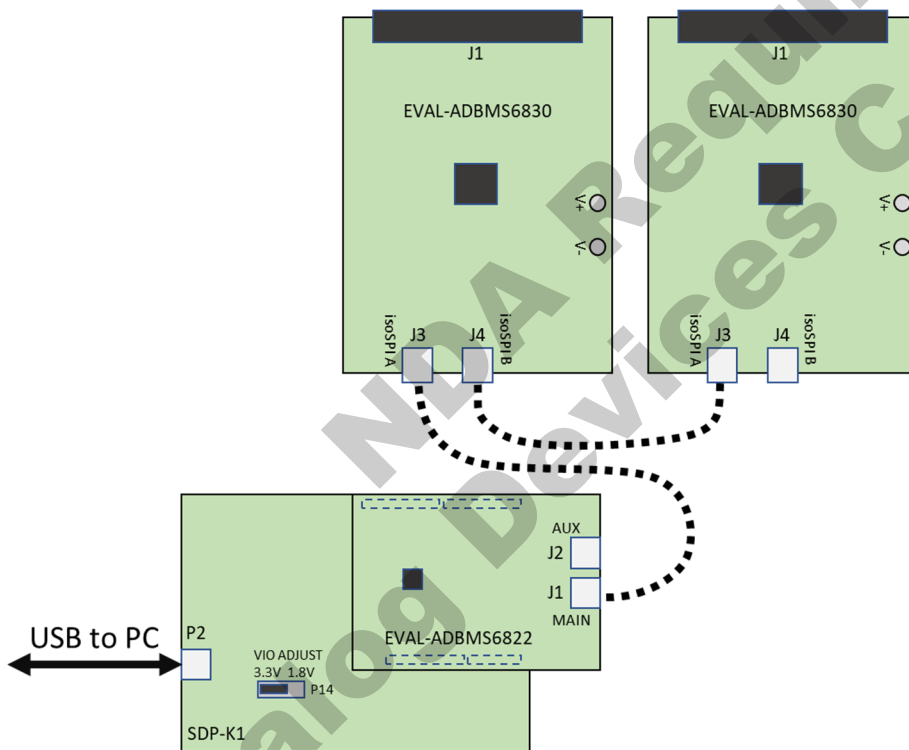


Figure 4. EVAL-ADBMS6830 Typical isoSPI Connection to the “Bottom” EVAL-ADBMS6830 in a Two-Board EVAL-ADBMS6830 Stack

EVAL-ADBMS6822 to EVAL-ADBMS6830 REVERSE isoSPI CONNECTION

A reverse isoSPI connection begins with the isoSPI Master connected to the last (or “top”) EVAL-ADBMS6830. Additional EVAL-ADBMS6830¹ boards can be daisy-chained onto the isoSPI bus. Communication begins from the last (or “top”) EVAL-ADBMS6830 then to the next “lower” EVAL-ADBMS6830 and, finally, to the first (or “bottom”) EVAL-ADBMS6830.

Figure 5 shows the following connections for two boards on a stack interfaced to a PC:

1. Connect the EVAL-ADBMS6822 Dual Master isoSPI demo board to the SDP-K1.
 - a. mate the EVAL-ADBMS6822 board directly onto the ‘shield’ headers of the SDP-K1.

- b. Set the P14 jumper of the SDP-K1 to the 3.3V position.
2. Connect a USB cable from the PC USB port to the SDP-K1 P2 connector.
3. Connect the EVAL-ADBMS6822 to the EVAL-ADBMS6830 in isoSPI mode. This EVAL-ADBMS6830 is the last (or “top”) board of a two-board stack.
 - a. Connect a 2-wire twisted-pair patch cable² from the EVAL-ADBMS6822 J1 MAIN DuraClik® connector to the “top” EVAL-ADBMS6830 J4 isoSPI B DuraClik® connector.
4. Connect or daisy-chain the EVAL-ADBMS6830 to another EVAL-ADBMS6830 in isoSPI mode. This EVAL-ADBMS6830 is the first (or “bottom”) board of a two-board stack. More EVAL-ADBMS6830 “lower” boards can be daisy-chained together in the same manner.
 - a. Connect a 2-wire twisted-pair patch cable² from the “top” EVAL-ADBMS6830 J3 isoSPI A DuraClik® connector to the next “lower” or “bottom” EVAL-ADBMS6830 J4 isoSPI B DuraClik® connector.
5. CAUTION! Prevent damage to the EVAL-ADBMS6830. Refer to Table 1 and confirm that the cell-voltage connections to the screw-terminal block matches the EVAL-ADBMS6830 J1 pinout.
 - a. Plug the screw-terminal block(s) into the J1 cell-voltage connector(s).
6. A Windows PC based Graphical User Interface is available. Refer to the ADBMS68xx_GUI Software Manual to properly setup the PC with the ADBMS68xx_GUI software to allow communication to the EVAL-ADBMS6830 boards. Refer to the Where To Get The ADBMS68xx_GUI? of this demo manual to request this software.

¹ DC3012A, Legacy DC3012A are compatible with EVAL-ADBMS6830.

² CAUTION! The 2-wire twisted-pair patch cable with the DuraClik® end plugs have 1mm thick center locking tabs on the wiring side that must be pressed down to release from the DuraClik® receptacles. Failure to do so may damage the cable and prevent board-to-board isoSPI communication.

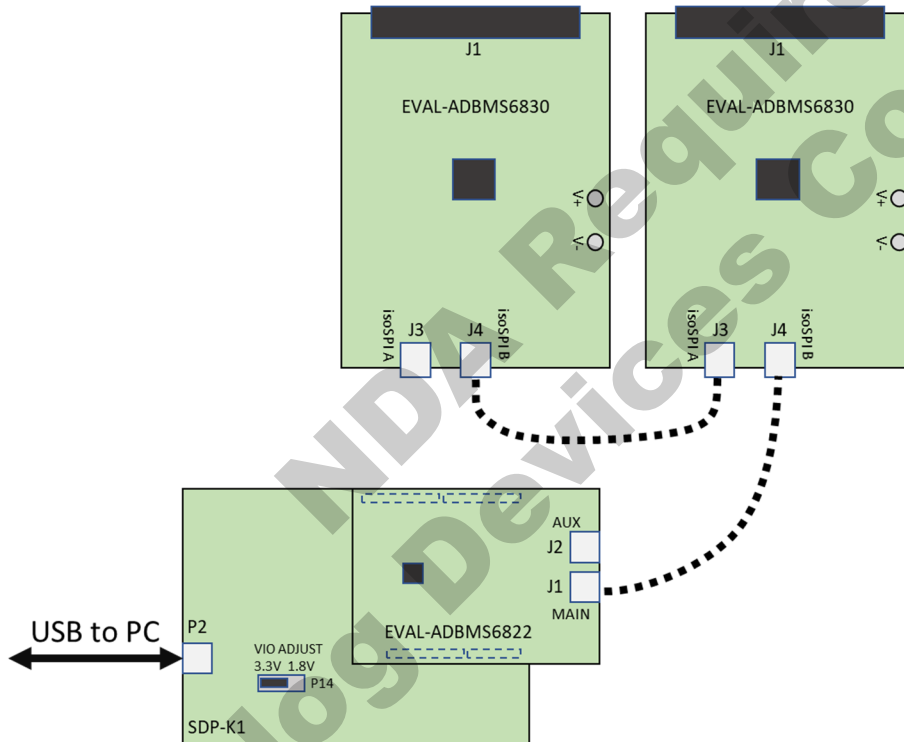


Figure 5. EVAL-ADBMS6830 Reverse isoSPI Connection to the “Top” EVAL-ADBMS6830 in a Two-Board EVAL-ADBMS6830 Stack

EVAL-ADBMS6822 to EVAL-ADBMS6830 REDUNDANT isoSPI CONNECTION

A redundant isoSPI connection begins with the primary (or “main”) isoSPI Master connected to the first (or “bottom”) EVAL-ADBMS6830 and has a backup auxiliary (or “aux”) isoSPI Master connected to the last (or “top”) EVAL-ADBMS6830. Additional EVAL-ADBMS6830¹ boards can be daisy-chained between the two isoSPI masters on the isoSPI bus. Primary (or “main”) communication begins from the first (or “bottom”) EVAL-ADBMS6830 then to the next “upper” EVAL-ADBMS6830 and, finally, to the last (or “top”) EVAL-ADBMS6830. The backup auxiliary (or “aux”) communication begins in the reverse direction to provide coverage when a possible isoSPI daisy-chain break occurs.

Figure 6 shows the following connections for two boards on a stack interfaced to a PC:

1. Connect the EVAL-ADBMS6822 Dual Master isoSPI demo board to the SDP-K1.
 - a. mate the EVAL-ADBMS6822 board directly onto the ‘shield’ headers of the SDP-K1.

- b. Set the P14 jumper of the SDP-K1 to the 3.3V position.
2. Connect a USB cable from the PC USB port to the SDP-K1 P2 connector.
3. Connect the EVAL-ADBMS6822 primary (or “main”) isoSPI master to the first (or “bottom”) EVAL-ADBMS6830 board of the stack.
 - a. Connect a 2-wire twisted-pair patch cable² from the EVAL-ADBMS6822 J1 MAIN DuraClik® connector to the “bottom” EVAL-ADBMS6830 J3 isoSPI A DuraClik® connector.
4. Connect or daisy-chain the EVAL-ADBMS6830 to another EVAL-ADBMS6830 in isoSPI mode. This EVAL-ADBMS6830 is the last (or “top”) board of a two-board stack. More EVAL-ADBMS6830 “upper” boards can be daisy-chained together in the same manner.
 - a. Connect a 2-wire twisted-pair patch cable² from the “bottom” EVAL-ADBMS6830 J4 isoSPI B DuraClik® connector to the next “upper” or “top” EVAL-ADBMS6830 J3 isoSPI A DuraClik® connector.
5. Connect the EVAL-ADBMS6822 auxiliary (or “aux”) isoSPI Master to the last (or “top”) EVAL-ADBMS6830 board of the stack.
 - a. Connect a 2-wire twisted-pair patch cable² from the EVAL-ADBMS6822 J2 AUX DuraClik® connector to the “top” EVAL-ADBMS6830 J4 isoSPI B DuraClik® connector.
6. CAUTION! Prevent damage to the EVAL-ADBMS6830. Refer to Table 1 and confirm that the cell-voltage connections to the screw-terminal block matches the EVAL-ADBMS6830 J1 pinout.
 - a. Plug the screw-terminal block(s) into the J1 cell-voltage connector(s).
7. A Windows PC based Graphical User Interface is available. Refer to the ADBMS68xx_GUI Software Manual to properly setup the PC with the ADBMS68xx_GUI software to allow communication to the EVAL-ADBMS6830 boards. Refer to the Where To Get The ADBMS68xx_GUI? of this demo manual to request this software.

¹ DC2848B, Legacy DC3012A are compatible with EVAL-ADBMS6830.

² CAUTION! The 2-wire twisted-pair patch cable with the DuraClik® end plugs have 1mm thick center locking tabs on the wiring side that must be pressed down to release from the DuraClik® receptacles. Failure to do so may damage the cable and prevent board-to-board isoSPI communication.

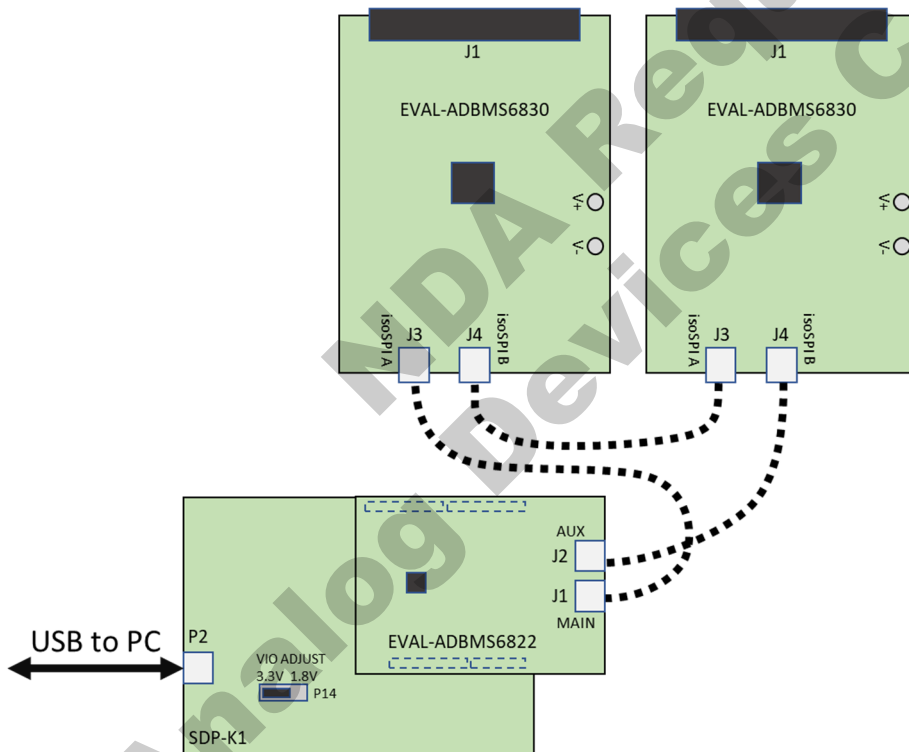


Figure 6. EVAL-ADBMS6830 Redundant isoSPI Connection to the “Bottom” EVAL-ADBMS6830 and “Top” EVAL-ADBMS6830 in a Two-Board EVAL-ADBMS6830 Stack

EVALUATION BOARD SOFTWARE

WHERE TO GET ADBMS68XX_GUI SOFTWARE?

Request the GUI software with the **ADI Software Request Form** from the following link:

- https://form.analog.com/form_pages/softwaremodules/SRF.aspx.
- Or search for “**Software Request Form**” at www.analog.com.

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NOTES



ESD Caution

ESD (electrostatic discharge) sensitive device. Charged devices and circuit boards can discharge without detection. Although this product features patented or proprietary protection circuitry, damage may occur on devices subjected to high energy ESD. Therefore, proper ESD precautions should be taken to avoid performance degradation or loss of functionality.

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